Fiscal Year:	FY 2016	**Task Last Updated:**	FY 06/10/2016
PI Name:	Matin, A.C. Ph.D.		
Project Title:	Spaceflight Effects on Bacterial Antibiotic Resistance and Its Genetic Basis “AntimicrobialSat”		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	(1) Developmental Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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City:	Stanford	**State:**	CA
Zip Code:	94305-5101	**Congressional District:**	14
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2008 SALMON--Astrobiology & FSB NNH08ZDA009O-SCMAFSB
Start Date:	07/01/2010	**End Date:**	04/30/2017
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:	Sato, Kevin	**Contact Phone:**	650-604-1104
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Flight Program:	Small Satellites		
Flight Assignment:	NOTE: End date changed to 04/30/2017 per F. Hernandez/ARC and NSSC information (Ed., 4/12/17) NOTE: End date changed to 3/31/2017 per PI and NSSC information (Ed., 6/13/16) NOTE: End date changed to 3/31/2016 per NSSC information (Ed., 1/20/16) NOTE: End date changed to 11/30/2015 per NSSC information and A. Chu/ARC (Ed., 12/16/14) NOTE: New end date is 11/30/2014 per NSSC information (Ed., 4/4/14) NOTE: New end date is 4/30/2014 per A. Chu/ARC and NSSC (Ed., 2/21/2013)		
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):			
Grant/Contract No.:	NNX10AM90A		
Performance Goal No.:			

Performance Goal Text:	
Task Description:	This project is concerned with the design, development, and testing of a nanosatellite payload, and the proposed biological experiment during spaceflight. The flight is now scheduled later in 2016. As in the past, the NASA and Stanford teams have worked closely on this project period to optimize the experimental conditions and ensure the success of the experiment in space. Astronauts during spaceflight are susceptible to bacterial infections; this is because microgravity of space compromises human immune system and as learned mostly by experiments conducted in microgravity simulation on Earth, it also makes bacteria more resistant to antibiotics. The EcAMSat experiment focuses on directly documenting alterations in antibiotic resistance in the space environment using a well-defined microbial system -- the bacterium, E. coli, and a widely-used antibiotic, gentamicin, against a uropathogenic strain of this bacterium. Fully autonomous microsystems like EcAMSat reduce the cost and obviate other limitations of human-tended space biological experiments. The Matin group has shown in Earth experiments, using simulated microgravity systems, that the increase in bacterial antibiotic resistance depends on their intact innate resistance, which is under the control of a master regulator gene called rpoS (3). Our recent findings have identified several proteins controlled by this gene that determine E. coli resistance to gentamicin (4). Future work to assess the role of these genes in space can be expected to provide further novel means of countering bacterial resistance in space. EcAMSat's 14 kg mass comprises provides state of the art technology to conduct preplanned experiments in space, e.g., a triple-junction solar cells, the experiment containment vessel, payload electronics subsystems, and bidirectional communication using a commercial WiFi-technology radio. Thus, experiments can be conducted during spaceflight and the bacterial sensitivity to the antibiotic in microgravity can be determined and transmitted to Earth. The experiments will determine if the rpoS gene is important to increased gentamicin resistance of E. coli in microgravity. If proven, countermeasures can be devised against this gene for the safety of astronauts. The work so far has completed preparations for spaceflight, i.e., equipment and experimental protocol have been finalized for the spaceflight. As soon as the flight is arranged, the microgravity research will begin. A paper reporting this accomplishment will be submitted for publication within a week or two. During the flight the data transmitted to Earth via WiFi will be analyzed by the Stanford and NASA groups; they will be interpreted in the light of several control experiments that have been done both at Stanford and at Moffett Field. References: 1. Principles of Clinical Medicine for Space Flight M. R. Barratt, S. L. Pool, Eds. (Springer, New York, ed. 1, 2008). 2. Ricco, A.J., Hines, J.W., Piccini, M., Parra, M., Timucin, L., Barker, V., Storment, C., Friedericks, C., Agasid, E., Beasley, C., Giovangrandi, L., Henschke, M., Kitts, C., Levine, L., Luzzi, E., Ly, D., Mas, I., Mclntyre, M., Oswell, D., Rasay, R., Ricks, R., Ronzano, K., Squires, D., Swaiss, G., Tucker, J., and Yost, B., "Autonomous genetic analysis system to study space effects on microorganisms: Results from orbit," Proc. Transducers 2007, IEEE: New York (2007); pp. 33-37. 3. Singh, R, and Matin, A C. In preparation 4. 1. J-H Wang, R Singh, M Benoit, M Keyhan, M Sylvester, M Hsieh, A Tathireddy, Y-J Hsieh, AC Matin. 2014. Sigma S-dependent antioxidant defense protects stationary phase Escherichia coli against the bactericidal antibiotic gentamicin. Antimicrob. Agents Chemother. 58(10): 5964-5975
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Antibiotic resistance in bacteria is becoming an extremely serious problem on Earth that will impact space travellers as well (the astronauts carry with them the Earth microflora). The research underway will provide a critical test for a hypothesis, which if confirmed, will show the way to greatly mitigate bacterial resistance to antibiotics in general. This hypothesis is based on a firm foundation provided by experiments conducted on Earth; since bacterial resistance increases under microgravity conditions, its testing in space provides an amplified situation where the difference made by the absence of the gene responsible can be seen more clearly. The innovative technology that is being developed is a generation of small molecules that can silence the gene in question, which can then either be used alone (since the resulting weakened bacteria can be more easily dealt with by the immune system) or in combination with existing antibiotics for effective treatment of infectious disease here on Earth as well as in space.
Task Progress:	Experimental conditions have been completed both in terms of the equipment and the experimental protocol for determining if the rpoS gene is responsible for the increased resistance to the antibiotic gentamicin of E. coli that causes urinary infection. If proven, countermeasures will be designed against this gene for the safety of astronauts.
Bibliography Type:	Description: (Last Updated: 03/27/2020)
Books/Book Chapters	Singh R, Matin AC. "Cellular response of Escherichia coli to microgravity and microgravity analogue culture." in "Effect of spaceflight and spaceflight analogue culture on human and microbial cells: Novel insights into disease mechanisms." Ed. C.A. Nickerson, N.R. Pellis, C.M. Ott. New York : Springer, 2016. p. 259-282. http://dx.doi.org/10.1007/978-1-4939-3277-1_13 , May-2016